

Benjamin Karachun

Biotechnology & Food Engineering
Scientific Computing
Laboratory & Data Analysis

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PROFILE

Biotechnology and Food Engineering student at the Technion, specializing in computational biotechnology. Trained in microbiology, molecular biology, analytical methods, and quantitative analysis. Builds scientific software for laboratory calculations and public datasets. Seeking a biotech internship or research-lab role.

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SCIENTIFIC & SOFTWARE PROJECTS

CFU Calculator Microbiology · Quantitative analysis · JavaScript [Open tool](#)

- Built an interactive CFU/mL calculator for serial-dilution plate counts, with technical and biological replicates, countability validation, uncertainty estimates, detection limits, and step-by-step calculations.
- Earned the highest laboratory bonus in the cohort after instructor evaluation.

Shoresh Python ETL · SQLite · WebAssembly [Open tool](#)

- Built a course-tailored alternative to Tzameret from public Israeli Ministry of Health datasets, supporting food and nutrient analysis, recipes, processing-retention calculations, comparisons, and label generation.
- Validated shared calculations side by side with the course teaching assistant against Tzameret, reproducing identical outputs for matched inputs; awarded 20 course bonus points.

Circuit Atlas Scientific computing · Neuroscience · AI [Open atlas](#)

Built a mouse brain connectivity platform with a 1,099 × 1,099 matrix and about 7.7 million Allen measurements. Added cohort comparisons, links to original evidence, and AI research modes that use matrix data or source records.

TechnionPrep Next.js · React · TypeScript [bfe.technionprep.com](#)

Built and deployed a semester-by-semester study hub that organizes course materials, summaries, and interactive tools for Technion Biotechnology and Food Engineering students.

MathPaster Chrome extension + web editor [mathpaster.com](#)

Built a MathLive-based WYSIWYG editor with autocomplete, matrix tools, history, and direct LaTeX insertion into AI chat interfaces. Equation data stays local to the browser.

02

EXPERIENCE

Python Tutor Technion Student Community

- Tutored undergraduates in the Technion's Python programming course.
- Built and maintained TechnionPrep's Python study hub for course materials and exercises.

Technology Consultant Hanan Shemesh Accounting

- Integrated the WhatsApp Business API to automate client requests and routine communication.
- Digitized legacy records and introduced a searchable operational workflow.

Web Developer Aviator.co.il

- Rebuilt the legacy WordPress/Elementor site as lean, responsive static HTML, CSS, and JavaScript while preserving its content and URLs.
- Maintained production infrastructure, managed updates and assets, and resolved operational issues.

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EDUCATION

Technion — Israel Institute of Technology
B.Sc. Biotechnology & Food Engineering
Expected 2027 · Haifa, Israel
Computational Biotechnology, Food Engineering & Health
Relevant coursework: Analytical Methods, Microbiology, Food Microbiology, Molecular Biotechnology, Biochemistry, and Metabolic Pathways

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RESEARCH INTERESTS

Antimicrobial resistance; *Staphylococcus aureus*; host-directed immune engineering and immune modulation; gene and cell editing; computational and AI-guided therapeutic design.

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LABORATORY

MICROBIOLOGY
Aseptic technique, media preparation and inoculation, streak plating, serial dilution, viable colony counting, Gram staining, microscopy, disc diffusion, zones of inhibition, and MIC assays

MOLECULAR BIOLOGY
PCR and gel electrophoresis

ANALYTICAL
Spectrophotometry, chromatography, and 96-well plate assays

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QUANTITATIVE & TECHNICAL

SCIENTIFIC ANALYSIS
Calibration and standard curves, concentration calculations, regression, uncertainty and error analysis, statistical comparisons

DATA & PROGRAMMING
Python, Pandas, NumPy, SQL, SQLite, Excel

WEB & RUNTIMES
TypeScript/JavaScript, React, HTML/CSS, WebAssembly, Pyodide

TOOLING & AI
Git, Docker, Linux/Bash; OpenAI, Gemini, and Claude APIs

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LANGUAGES

Hebrew (native) · **English** (fluent) · **Russian** (fluent)